EXECUTIVE AUDIT

Disclaimer: AI generated presentation based on prompts for the AI exercise for Decision Analytics course.

Decision Analytics prompt quality

Eight prompts for Brussels student housing investment

Comparative assessment and ten recommendations

29 October 2026

Group 12 offers the clearest starting point

Recommended approach: combine the best controls across prompts

Foundation: Group 12

Compact workbook architecture, explicit financial perspectives, named cells and legal verification flags.

Essential additions: original and Project XL

Preserve negative outcomes and a zero-NPV no-investment option. Add a rent-demand relationship.

Selective additions: Group 7Wand Group10

Retain detailed validation and affordability controls. Remove repetition and repair Group10’s negative-input rule.

Assessment of expected reliability. Actual investment-output quality remains untested.

Evidence and audit boundaries

Eight prompts reviewed. No generated investment models or reports supplied.

Prompt	Words*	Distinctive request
Original	1,322	Improve an existing workbook using a class example
Project XL	944	Integrated model and executive report in seven minutes
W	21,408	Workbook, PDF, slides and ZIP with extensive methods
Group 13	401	Property ranking and 50,000 simulations per property
Group 12	469	Compact workbook specification with legal flags
Maite	164	Agile iterations and classification of inputs
Group 14	284	Basic workbook and analyses in seven minutes
Group10	1,999	Ixelles feasibility, affordability and due diligence

*Whitespace word counts of saved text files. The earlier .txt conversions demonstrate transcription only.

Prompt design scorecard

Six equally weighted dimensions, each scored 1–5. Maximum 30.

Prompt	Clarity	Evidence	Rigor	Feasibility	Checks	Economy	Total
Group 12	5	5	3	4	4	5	26
Original	5	4	4	3	4	4	24
Project XL	4	4	4	2	4	4	22
Group10	4	5	2	3	5	3	22
W	3	5	5	2	5	1	21
Group 13	4	4	4	2	2	4	20
Group 14	3	4	3	2	2	4	18
Maite	2	2	2	3	1	5	15

Auditor judgments, not measured model performance. Critical defects override total scores.

Expected outputs: strongest detailed briefs

Predictions from the instructions. No investment outputs have been benchmarked.

Prompt	Likely output strengths	Main quality risk
Original	Course-aligned workbook with sensitivity, optimization and risk analysis	Seven-minute budget and missing reference workbooks constrain execution
Project XL	Model plus concise report with rent affecting demand	Research, optimization and reporting compete for seven minutes
W	Extensive documentation, financial controls and broad analytical coverage	Repeated instructions and many artifacts increase omission and inconsistency risk
Group10	Strong feasibility, affordability, timeline and financing analysis	Negative-input rule may bias results. Course optimization and Monte Carlo unclear

Coverage in a prompt creates a requirement. It does not establish that the resulting workbook works.

Expected outputs: compact briefs need precise contracts

Short prompts can execute well when they define the right constraints.

Prompt	Likely output strengths	Main quality risk
Group 12	Auditable annual workbook, separate project/equity returns and legal flags	Optimization execution, probability assumptions and demand response need detail
Group 13	Commercial property ranking with timelines and downside analysis	No defined risk-adjusted objective or reproducibility protocol for 50,000 trials
Group 14	Basic investment model with terminal-value comparison	“3 different maps” is ambiguous. Solver and simulation lack specifications
Maite	Useful problem framing, visuals and input classification	No horizon, KPI contract, simulation protocol or testable definition of completion

Group 12 defines a Solver setup, which is narrower than a verified optimized result.

Three instruction defects threaten decision quality

These are analytical and execution risks. The supplied text shows no destructive filesystem commands.

Group10: “Don't take negative inputs into account”

Could suppress losses or initial outflows. Restrict validation to physically impossible inputs and retain negative financial outcomes.

Original, Project XL, Group 14: seven minutes

A deadline alone cannot ensure verified research and working analyses. Define a minimum deliverable and a stopping rule.

Maite and Group 14: ambiguous requirements

Specify what each iteration must improve. Replace “maps” with named worksheets, methods, outputs and pass criteria.

W’s repeated content adds avoidable execution risk

The saved prompt contains four master headings and substantial repeated sections.

21,408

words in the supplied Group 7Wprompt

Other seven prompts: 164–1,999 words each

Retain Group 7Was a requirements catalogue. Extract each requirement once into a staged execution brief.

Length alone does not establish poor quality. Repetition and weak prioritization create the avoidable burden.

Recommendations 1–3: scope and execution

Executive priority: a coherent specification that the available tools can complete

01 One authoritative decision brief

Define project type, investor objective, affordability policy and required artifacts. Keep source documents separate.

Owner: project lead. Acceptance: one agreed scope with no conflicting requirements.

02 A single copy of every requirement

Remove W's repeated blocks. Replace role stacking and superlatives with explicit outputs and acceptance criteria.

Owner: prompt owner. Acceptance: every requirement appears once and has a clear priority.

03 Staged delivery with saved checkpoints

Separate evidence collection, baseline model, analyses and reporting. Set a minimum verified deliverable for a time limit.

Owner: delivery lead. Acceptance: each stage saves a usable artifact and records incomplete work.

Recommendations 4–6: financial and evidence controls

Executive priority: prevent a polished model from producing a biased decision

04 Preserved losses and a no-investment option

Replace Group10’s negative-input rule. Retain negative NPV and cash flows. Compare construction with an NPV of zero.

Owner: finance reviewer. Acceptance: a loss-making fixture produces a loss and can recommend rejection.

05 Evidence records for every material assumption

Record value, unit, source, date, confidence and status. Treat local legal applicability as unresolved until verified.

Owner: research lead. Acceptance: every legal claim has a source and applicability status.

06 An explicit financial model contract

Define timing, signs, rent-demand logic and terminal value. Separate project cash flow from financing and equity returns.

Owner: model lead. Acceptance: independent checks reconcile funding, debt and discounted cash flow.

Recommendations 7–10: implementation and proof

Executive priority: reproducible outputs and truthful completion claims

07 **Declared tool capabilities and safe file handling**

State native Excel versus equivalent calculations. Save new versions in an output folder and review any macro code.

Owner: execution lead. Acceptance: no source overwrite and no unexecuted Solver claim.

08 **A justified uncertainty protocol**

Specify distributions, dependencies and seed. Increase trials until the decision metrics are sufficiently stable.

Owner: risk lead. Acceptance: reproducible trials and a documented convergence check.

09 **Measurable acceptance tests**

Test input propagation, losses, zero occupancy, quota rounding and debt reconciliation. Verify charts and file opening.

Owner: independent reviewer. Acceptance: pass/fail evidence for every critical check.

10 **A controlled comparison of generated outputs**

Run all candidates on identical inputs and tool settings. Score artifacts blindly and record time, cost and failures.

Owner: audit lead. Acceptance: repeated runs and a documented basis for selecting a winner.

A consolidated prompt needs six clear sections

Recommended structure for the next version

Section	Required instruction
1. Decision	Choose project type and alternatives. Define the investor objective and affordability policy.
2. Inputs and evidence	List attached files. Separate source content from instructions. Record missing data and status.
3. Financial contract	Define horizon, units, signs, cash-flow perspectives, demand response and terminal value.
4. Analysis and tools	Specify sensitivity, feasible optimization and simulation protocol with tool-specific fallbacks.
5. Validation	Require numerical tests, independent reconciliations and honest completion disclosures.
6. Deliverables	Name the files, stage priorities, safe save location and minimum acceptable completion.

Use Group 12 for structure, the original for downside integrity, and selected detail from the other prompts.

A benchmark must test the artifacts

Proposed evaluation protocol. No benchmark runs have been executed.

Control	Proposed test
Common environment	Same model version, tools, source workbook, class materials, evidence pack and compute budget
Repeated runs	At least three runs per prompt. Retain failures and report variation rather than only the best run
Independent scoring	Blind review of saved artifacts using the same course rubric and financial invariants
Quality measures	Formula correctness, input propagation, feasible optimization, uncertainty logic and source support
Delivery measures	Working files, required content completed, elapsed time, resource cost and manual repair effort
Critical failure gate	Reject output with suppressed losses, fabricated results, unsupported legal constraints or broken cash flows

Course scoring: model 8 points, sensitivity 4, optimization 4, Monte Carlo 4. Use prompt scores separately.

Numerical fixtures expose common failure modes

Examples for the proposed benchmark, independent of Brussels market assumptions

Fixture	Expected behavior	Defect detected
Initial investment –€100 Year 1 inflow €110, rate 10%	NPV = €0 IRR = 10%	Period-0 omission or incorrect signs
Zero occupancy and positive fixed costs	Rental income = €0 Operating cash flow remains negative	Suppressed losses or revenue at full capacity
31 rooms, project policy requires ≥10% affordable	At least 4 affordable rooms 3 / 31 = 9.68%, which fails	Incorrect integer rounding or quota calculation
Higher rent in a model with price-sensitive demand	Demand falls under the stated function Revenue may rise or fall	Fixed occupancy despite mandatory demand response
€100 opening debt, no draw, €10 principal repayment	Closing debt = €90 Interest remains separate	Debt imbalance or double-counted financing

Illustrative tests. The 10% quota is a test policy, not a statement of Brussels law.

The next version can retain each prompt’s best contribution

Leadership decision: consolidate, repair, then benchmark

Prompt	Contribution to retain	Change before use
Group 12	Compact model architecture and legal flags	Add execution evidence and demand response
Original	Negative outcomes and no-investment choice	Replace deadline pressure with staged delivery
Project XL	Rent-demand logic and genuine optimization	Separate research and model/report stages
W	Formula, validation and implementation detail	Remove repeated copies and select core requirements
Group 13	Property economics and timing	Define risk objective and simulation precision
Maite	Input classification and iteration	Add acceptance gates and deliverable specification
Group 14	Terminal-sale comparison and local costs	Clarify worksheets, methods and tests
Group10	Feasibility and affordability gates	Repair negative-input rule and require course analyses

Audit basis and limitations

Prompt requirements describe intent. Artifact tests establish delivered quality.

Method

Text inspection, measured word counts, six-dimension ordinal scoring and evidence-linked recommendations. Scores reflect auditor judgment.

Best-practice references

OpenAI: Prompt engineering

OpenAI: Reasoning best practices

No generated investment outputs, actual runtime tests or Brussels legal verification form part of this audit.

Sources retrieved 29 October 2026. Primary audit evidence: the eight supplied prompts.